Rohit Udaiwal

AI Engineer
udaiwalrohit@gmail.com
linkedin.com/in/rohit-udaiwal
+91 8529337644
rohit.udaiwal.com
Bangalore, India

EXPERIENCE

RISA Labs
March 2026 - Present

Senior Member of Technical Staff — Bangalore, India

- **Knowledge Graphs & Clinical Trial Eligibility:** Building knowledge graphs for clinical trial eligibility screening using a funnel-based approach: filtering patients criterion by criterion through structured Cypher queries and narrative queries over patient medical records to answer clinical eligibility questions.
- **RPA & Portal Integration:** Engineered payer-portal workflows for US healthcare pharmacy prior authorization (PA), eliminating up to 60% of manual form submissions and progressing toward no-touch PA submissions.
- **Medical Necessity Agent:** Built an agent that reasons over clinical charts against medical-necessity criteria to answer PA questionnaires and predict approval likelihood, using MCP tools for CMS/OpenFDA and a custom PA skill on the Claude Agent SDK.
- **PA Status Voice Agent:** Built a voice agent to retrieve real-time PA status from PBMs through tool calls, with IVR navigation, live call listening, human takeover, and real-time status updates.

Finfactor Technology
June 2025 - October 2025

AI Engineer — Pune, India

- **Intelligent Customer Support:** Developed a centralized support chatbot using Flask, LangChain, retrieval-augmented generation (RAG), and Model Context Protocol to route queries by request type.
- **Transaction Classification:** Worked on a bank-statement narration classifier using Quarkus, React, Next.js, LLMs, BERT-based NLP, and vLLM for personal finance and lending decisions.
- **Financial AI:** Led development of machine learning solutions using NLP and large language models for real-time transaction analysis and financial decision-making.

JPMorgan Chase & Co.
August 2023 - May 2025

Software Development Engineer - Wealth Management — Mumbai, India

- **Scalable Data Pipelines:** Developed 50+ multi-region data pipelines to collect, process, and store diverse datasets in the central Wealth Management data lake.
- **Modernization:** Built data-processing workflows with Python, PySpark, Spring Boot, AWS, and Docker; created an enterprise Spring Boot application adopted by 100+ teams.
- **CI/CD Automation:** Developed Jenkins and Docker CI/CD pipelines that reduced deployment times by 70%.

TECHNICAL SKILLS

Languages & Querying: Python, C++, Java, JavaScript, C, MySQL, Cypher

Frameworks & Libraries: Flask, Quarkus, Spring Boot, FastAPI, Django, React, Next.js, Spark, PyTorch

AI Tools & Techniques: Knowledge graphs, clinical eligibility screening, voice agents, vLLM, PEFT, LoRA, RAG, LangChain, MCP, Claude Agent SDK

Platforms & Services: AWS, Kubernetes, Docker, Kafka, Terraform, Jenkins, Linux, Dynatrace, Grafana, PostHog, Snowflake

PUBLICATION

Deep Learning based Automatic Modulation Classification
IEEE SPCOM '24

IIT Guwahati | May 2023 - July 2023

Developed a quad-stream architecture combining LSTMs and attention for RF-signal classification, achieving 99% accuracy in the reported evaluation. Implemented custom attention layers in TensorFlow and Keras, with applications in cognitive radio, spectrum monitoring, and civilian and defense communications.

EDUCATION

Indian Institute of Technology (IIT), Guwahati
June 2023

B.Tech in Electronics and Electrical Engineering | CGPA: 8.38/10.0

HONORS & AWARDS

Codeforces Expert - maximum rating 1817 | AWS Certified Solutions Architect - Associate (2023) | JEE Main 2019 - top 0.3% of 1.3 million participants | JEE Advanced 2019 - rank 4285 | NTSE - qualified for national level; 140/150 marks